CCIT Report #570
January 2006

Classification and Segmentation of Textures based on Statistical Models of Wavelet Transforms

R. Mittelman and M. Porat
Department of Electrical Engineering
Technion-Israel Institute of Technology
Haifa 32000, Israel

Abstract

In this work we introduce new multiresolution statistical models for texture classification and segmentation. We develop features for wavelet-based texture classification using statistical modeling based on a Gaussian scale mixture (GSM), and present a new feature weighting scheme suitable for histogram features. Texture segmentation is performed using Markov random fields and variational level-set frameworks. Multi-scale variational level-set texture segmentation is introduced, which in contrast to presently available segmentation schemes, employs a coarse to fine strategy. The new scheme is based on the dual tree complex wavelet transform and hidden Markov trees statistical modeling. The proposed scheme achieves better classification results than those reported in the literature for the same textures and tasks. Classification and segmentation examples are presented and discussed. The segmentation results indicate that the new scheme is superior to presently available methods, in particular when segmenting non-convex shapes, since the coarse to fine strategy facilitates the dependence on the rigidity of the contour. Our conclusion is that the proposed approach could improve presently available tools for classification and segmentation of textures.

Corresponding Author:

Roni Mittelman,
Department of Electrical Engineering,
Technion, Israel
Email: rm@vision.technion.ac.il